

Chapter 1

Section 7 - Discrete Probability

1.1 Section 7.1 - Intro to Probability

1.1.1 Event and Sample Space

Definition 1 (Event and Sample Space)

Let S be a finite, non-empty set called a **sample space** and a set $E \subseteq S$ called the **event space**. The probability of E occurring is defined as

$$P(E) = \frac{|E|}{|S|}$$

Example 1

Let's do some examples

1. Determine the probability of rolling the following numbers when 2 dice are rolled
 - (a) 12

S = possible rolls of 2 dice, $|S| = 36$

E = rolls that give a 12, $|E| = 1$

$$P(E) = \frac{|E|}{|S|}$$
$$= \boxed{\frac{1}{36}}$$

(b) 2

S = possible rolls of 2 dice, $|S| = 36$

E = rolls that give a 2, $|E| = 1$

$$P(E) = \frac{|E|}{|S|}$$
$$= \boxed{\frac{1}{36}}$$

(c) 3

S = possible rolls of 2 dice, $|S| = 36$

E = rolls that give a 3, $|E| = 2, (1 + 2), (2 + 1)$

$$P(E) = \frac{|E|}{|S|}$$
$$= \frac{2}{36}$$
$$= \boxed{\frac{1}{18}}$$

(d) 4

$$\begin{aligned}
 |S| &= 36 \\
 E &= \{(1, 3), (2, 2), (3, 1)\} \\
 |E| &= 3 \\
 P(E) &= \frac{3}{36} \\
 &= \boxed{\frac{1}{12}}
 \end{aligned}$$

This is just

Sum of dice	Probability
2	$\frac{1}{36}$
3	$\frac{2}{36}$
4	$\frac{3}{36}$
5	$\frac{4}{36}$
6	$\frac{5}{36}$
7	$\frac{6}{36}$
8	$\frac{5}{36}$
9	$\frac{4}{36}$
10	$\frac{3}{36}$
11	$\frac{2}{36}$
12	$\frac{1}{36}$

2. Determine the probability of rolling a 2 when a single die is rolled

$$\begin{aligned}
 S &= \{1, 2, 3, 4, 5, 6\} \\
 E &= \{2\} \\
 P(\text{rolling } 2) &= \boxed{\frac{1}{6}}
 \end{aligned}$$

3. Determine the probability of not rolling a 2 when a die is rolled

$$\begin{aligned}
 S &= \{1, 2, 3, 4, 5, 6\} \\
 E &= S - \{2\} \\
 &= \{1, 3, 4, 5, 6\} \\
 P(\text{not rolling } 2) &= \boxed{\frac{5}{6}}
 \end{aligned}$$

or

$$\begin{aligned}
 P(\text{not rolling } 2) + P(\text{rolling } 2) &= 1 \\
 P(\text{not rolling } 2) &= 1 - P(\text{rolling } 2) \\
 &= 1 - \frac{1}{6} \\
 &= \boxed{\frac{5}{6}}
 \end{aligned}$$

4. A card is chosen from a standard 52 deck. Determine the probability that the card is
- (a) An ace

$$\begin{aligned}
 |S| &= 52 \\
 |E| &= 4 \\
 P(\text{ace}) &= \frac{4}{52} \\
 &= \boxed{\frac{1}{13}}
 \end{aligned}$$

- (b) An spade

$$\begin{aligned}
 |S| &= 52 \\
 |E| &= 13 \\
 P(\text{spade}) &= \frac{13}{52} \\
 &= \boxed{\frac{1}{4}}
 \end{aligned}$$

(c) A spade or an ace

$$\begin{aligned}
 |S| &= 52 \\
 |E| &= |\text{Ace} \cup \text{Spade}| \\
 &= |\text{Ace}| + |\text{Spade}| - |\text{Ace} \cap \text{Spade}| \\
 &= 4 + 13 - 1 \\
 &= 16
 \end{aligned}$$

$$\begin{aligned}
 P(\text{ace or spade}) &= \frac{16}{52} \\
 &= \boxed{\frac{4}{13}}
 \end{aligned}$$

1.1.2 Probability with Multiple Subsets (OR)

From these examples we see that if $E_1, E_2 \subseteq S$, then

$$\begin{aligned}
 P(E_1 \cup E_2) &= \frac{|E_1 \cup E_2|}{|S|} \\
 &= \frac{|E_1| + |E_2| - |E_1 \cap E_2|}{|S|}
 \end{aligned}$$

1.2 Section 7.2 - Probability Theory

We didn't actually go over much

1.2.1 Event Independence

Definition 2 (Event Independence)

If H and Q are events, we say H and Q are **independent** if and only if

$$P(H \cap Q) = P(H) \cdot P(Q)$$

Example 2

A card is selected from a standard 52 deck. Determine the probability that

1. The card is a heart

$$\begin{aligned} P(\heartsuit) &= \frac{\# \text{ hearts}}{\# \text{ cards}} \\ &= \frac{13}{52} \\ &= \boxed{\frac{1}{4}} \end{aligned}$$

2. The card is a queen

$$\begin{aligned} P(\text{Queen}) &= \frac{\# \text{ queens}}{\# \text{ cards}} \\ &= \frac{4}{52} \\ &= \boxed{\frac{1}{13}} \end{aligned}$$

3. The card is a heart and a queen

$$\begin{aligned} P(\heartsuit \text{ and queen}) &= \frac{\# \text{ heart and queen}}{\# \text{ cards}} \\ &= \frac{1}{52} \end{aligned}$$

Notice that

$$P(\heartsuit \text{ and queen}) = P(\heartsuit) \cdot P(\text{queen})$$

This means that H and Q are **independent**

I would put the examples leading up to the derivation of this next thing but I'm too lazy

1.2.2 Binomial Probability

Definition 3 (Binomial Probability)

Given a limited event (ie 2 outcomes) with probability p , Then

$$P(k \text{ successes in } n \text{ trials}) = \binom{n}{k} p^k \cdot (1-p)^{n-k}$$

Example 3

Let's do some examples

A biased coin has probability of 0.51 for landing heads when flipped. Determine the probability of the coin landing on heads exactly 6 times when the coin is flipped 10 times

$$\begin{aligned} p &= 0.51 \\ k &= 6 && \text{(successes)} \\ n &= 10 \end{aligned}$$

$$\begin{aligned} P(6 \text{ successes in } 10 \text{ trials}) &= \binom{10}{6} (0.51)^6 \cdot (1-0.51)^{10-6} \\ &\approx \boxed{0.213} \end{aligned}$$

Example 4

Let's do another

Determine the probability of rolling a 12 **at least once** when a pair of dice is rolled 20 times.

Since we have "at least" let's do the opposite

$$P(\text{never rolling 12 out of 20}) = \binom{20}{0} \left(\frac{1}{36}\right)^0 \cdot \left(\frac{35}{36}\right)^{20} \\ \approx 0.569\dots$$

$$P(\text{rolling 12 at least once out of 20}) = 1 - P(\text{never rolling 12 out of 20}) \\ \approx 1 - 0.569 \\ \approx \boxed{0.431}$$

1.4 My Section